

1.1 Classwork: Frequency Tables — Markscheme

1. [Maximum mark: 6]

Raw data as printed (24 replies), with the counts:

bus subway walk car subway walk bike subway
walk bus subway subway car walk subway bus
subway bike walk bus car subway walk bus

How you travel	walk	bus	subway	car	bike	Total
Frequency	6	5	8	3	2	24

- (a) Tally marks recorded for all five categories **A1**

Frequencies 6, 5, 8, 3, 2 all correct **A1**

The tally A1 is for a tally that is used and that agrees with the candidate's own frequency column, however the strokes are grouped; the five-bar gate is not required. The frequency A1 requires all five values correct. Accept a completed frequency column with the tally column left empty for the second A1 only.

- (b) Total = $6 + 5 + 8 + 3 + 2 = 24$ **A1**

Accept 24 stated with no working, or read from the stem.

- (c) Mode is *subway*. **A1**

Accept "the subway" or "subway, 8 students". Do not accept "8" alone — the mode of a categorical variable is the category, not its frequency.

FT: award A1 for the category with the largest frequency in the candidate's own table from (a).

- (d) $\frac{6}{24} = \frac{1}{4} = 25\%$ **A1**

Accept any equivalent fraction, decimal or percentage; simplification is not required. FT: award A1 for the candidate's own walk frequency divided by the candidate's own total.

- (e) It is categorical, because the replies are names of groups (words), not numbers that can be measured or counted. **R1**

R1 is earned by any answer that identifies the variable as categorical and gives a reason of the right kind — the data are words / labels / groups rather than numerical measurements. "Categorical" with no reason earns nothing.

2. [Maximum mark: 6]

Raw data as printed (25 replies), with the counts:

1	2	0	3	1
1	2	4	1	2
1	3	0	2	1
1	2	3	1	2
1	4	0	2	3

Number of siblings	0	1	2	3	4	Total
Frequency	3	9	7	4	2	25

(a) Tally marks recorded for all five values A1

Frequencies 3, 9, 7, 4, 2 all correct A1

The tally A1 is for a tally that is used and that agrees with the candidate's own frequency column, however the strokes are grouped. The frequency A1 requires all five values correct. Accept a completed frequency column with the tally column left empty for the second A1 only.

(b) Total = $3 + 9 + 7 + 4 + 2 = 25$ A1

Accept 25 stated with no working, or read from the stem.

(c) Mode is 1. A1

Accept "1 sibling" or "one brother or sister". Do not accept "9" alone — that is the frequency, not the mode.

FT: award A1 for the value with the largest frequency in the candidate's own table from (a).

(d) At least 2 means 2, 3 or 4: $7 + 4 + 2 = 13$ students. A1

A1 is for the value 13, reached by any correct route — adding the frequencies for 2, 3 and 4, counting directly from the raw list, or subtracting from the total ($25 - 3 - 9 = 13$).

A candidate who reads "at least 2" as "more than 2" obtains 6; this earns no mark, since the whole mark is for the count. FT: award A1 for the sum of the candidate's own frequencies for 2, 3 and 4.

(e) It is discrete, because the number of brothers and sisters is counted and can only take separate whole-number values — you cannot have 2.4 siblings. R1

R1 is earned by any answer that says the variable is discrete and gives a reason of the right kind — the values are counted, are whole numbers, or cannot take values in between. "Discrete" with no reason earns nothing.